Causality

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Outline

Cause-and-effect

Potential outcomes and counterfactuals

Observational studies

From prediction to cause and effect

So far: supervised learning. Given X, what Y should we expect?

Houses with a fireplace sell for more. prediction

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Now: causal inference. If we *change* X, how does Y change?

If you *add* a fireplace, will your house sell for more? — cause and effect

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A model can predict Y very well and still give the wrong answer to a causal question.

And causal questions show up EVERYWHERE!

One intervention, opposite predicted effects

Prop 40: a one-time 5% tax on billionaires' net worth, mostly for health care. On the ballot November 3, 2026.

For:

> *"We are facing literally a collapse of our healthcare system here in California and elsewhere."*
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Against:

> *"Wealth is movable, and it shops for the state with the lowest taxes."* — Gov. Gavin Newsom, 2026

> *"There will be no billionaires left in California."* — Chamath Palihapitiya, 2025

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Same policy, opposite predicted effects. **How would we ever know who is right?**

Same data, opposite conclusions

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Both sides are looking at the same state, the same law, and the same years.

The disagreement is about the counterfactual: what would employment have been *without* the law?

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"I do agree that the three sites were obliterated. . . It set the program back, I think, a couple years."
— Sen. Lindsey Graham, 2025

Causal claims in science and technology

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Each is a claim about cause and effect. Which ones does the evidence actually support?

Cause-and-effect $\iff$ policy impacts

These two paradigms are one and the same!

Cause is a statement of something being manipulated or changed
Effect is a measure of the change in an outcome of interest

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Cause is a statement of something being manipulated or changed
Effect is a measure of the change in an outcome of interest

"Cause" is the same as a policy introduction or change

"Effect" is the unique, independent measurement of how the cause modulated some other part of our system

Cause-and-effect: policy decisions as cause

We see causes all of the time

- The federal government increases the minimum wage
- The FAA mandates face coverings on planes
- The percentage of vaccinated individuals exceeds 50% of population
- It's windy on an Austin afternoon
- Sweetgreen decreases its Kale caesar salad price by $2
- States implement stay-at-home orders during the pandemic

Cause-and-effect: varying features as cause

Causes might also just be variation in a population

- Are dog-lovers nicer people?
- Does race affect hiring decisions?
- Does age affect COVID-19 mortality?
- Is there a "gender-gap" in salary?

The **difficult** question is, what about the effects?

What outcomes do we look at?

How do we measure them?

Are there other variables that might affect the outcomes *and* the causes?

The **difficult** question is, what about the effects?

What outcomes? (**lower class unemployment rate, income, . . .**)

How to measure? (**government data, surveys, . . .**)

Other variables? (**current economic conditions, differences among states, . . .**)

Sweetgreen salad price increase

The **difficult** question is, what about the effects?

- What outcomes? (**revenue, count of kale caesars sold, . . .**)
- How to measure? (**financial data, . . .**)
- Other variables? (**seasonality, temperature, weather, . . .**)

Racial discrimination in hiring?

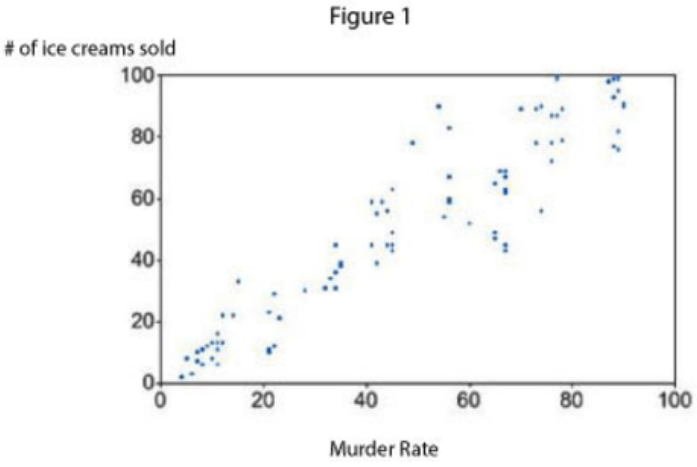
The **difficult** question is, what about the effects?

What outcomes? (**whether a job applicant receives a callback**)

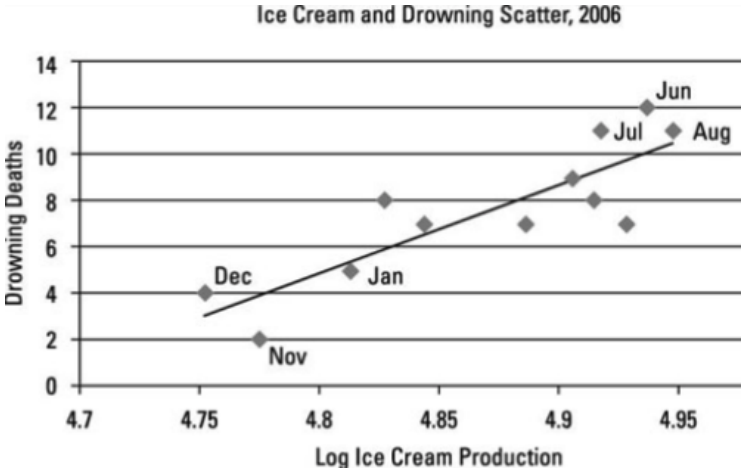
How to measure? (**follows directly from above**)

Other variables? (**other resume characteristics, GPA, university, . . .**)

Ice cream and NYC murder rate



Ice cream and drownings



To sum up cause-and-effect

Challenges are related to both the **system of study** and **ability to gather the right data**.

With data in hand, you can start to formulate hypotheses and test them.

There might be lurking variables driving an underlying relationship (ice cream). Only an expert (you!) can identify those and take them into account.

Let's formalize these ideas with some basic notation

Units of study indexed by i (**states, stores, customers, people**)

Cause aka treatment aka policy for unit i: z_i

$$z_i = \begin{cases} 0 & \text{not treated (control)} \\ 1 & \text{treated} \end{cases}$$

Outcome for each i as Y_i, written as a function of the cause:

$$Y_i(z_i) \leftarrow \text{potential outcome for unit } i$$

With this notation, we have the building blocks to talk about policy effects!

Example: COVID-19 lockdowns

How did state lockdowns at the beginning of 2020 affect the spread of the virus?

What are the potential outcomes?

Example: COVID-19 lockdowns

Let's set up the structure of this problem. Let i denote a state:

$$i \in \{\text{New York, California, Florida, Texas, South Dakota, } \dots\}$$

First, we define z_i:

$$z_i = \begin{cases} 0 & \text{no lockdown} \\ 1 & \text{lockdown} \end{cases}$$

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Second, we define our outcome: Y_i = cases per capita (in state i) after lockdown or no lockdown.

Example: COVID-19 lockdowns

A brief aside: Defining exactly what the treatment z_i is very hard! It could be a combination of many factors.

- Masking
- Bar and restaurant closures
- School closures
- Curfews
- Limits to exercise
- Retail store closures

Q: How would you define z_i?

Organizing our data: The Science Table

i **(state)**	z_i **(lockdown)**	$Y_i(0)$	$Y_i(1)$
New York			
Florida			
California			
Texas			
South Dakota			
Illinois			
⋮	⋮	⋮	⋮

Organizing our data: The Science Table

i (state)	z_i (lockdown)	$Y_i(0)$	$Y_i(1)$
New York	1		
Florida	0		
California	1		
Texas	0		
South Dakota	0		
Illinois	1		
⋮	⋮	⋮	⋮

Organizing our data: The Science Table (CDC, cases/100k)

i (state)	z_i (lockdown)	$Y_i(0)$	$Y_i(1)$
New York	1		.0034
Florida	0	.007	
California	1		.0014
Texas	0	.004	
South Dakota	0	.0028	
Illinois	1		.002
⋮	⋮	⋮	⋮

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California	1	??	.0014
Texas	0	.004	??
South Dakota	0	.0028	??
Illinois	1	??	.002
⋮	⋮	⋮	⋮

What is the ideal scenario?

We are able to know both of the potential outcomes for each state!

Y_{NY}  and Y_{NY} 

Defining a causal effect for NY

We can define the **causal effect** of the "lockdown treatment" as the difference between the two potential outcomes.

$$\tau_{\text{NY}} = Y_{\text{NY}}\left(\text{  }\right) - Y_{\text{NY}}\left(\text{  }\right)$$

Or written more generally:

$$\tau_i = Y_i(1) - Y_i(0)$$

The fundamental problem of causal inference

We only observe one of the two potential outcomes for each unit.

- **Economics of COVID policy**: a state either locks down or doesn't
- **Drug trials**: an individual either receives the medicine or the placebo
- **Gender wage gap**: a person is either male or female

The unknown outcomes are called the **missing potential outcomes** or **counterfactuals**.
This is what makes causality a nontrivial task... it is a missing data problem.

The potential outcomes framework is **always** our starting point.

Note: Randomized controlled trials (RCTs) are the **gold standard**—when treatment is randomly assigned, treatment and control groups are comparable, and causal effects are straightforward to estimate.

From potential outcomes to observational studies

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Note: Randomized controlled trials (RCTs) are the **gold standard**—when treatment is randomly assigned, treatment and control groups are comparable, and causal effects are straightforward to estimate.

But in many settings we have **observational data**: treatment is *not* randomized, it is simply observed.
"I observe states lockdown and not lockdown"
"I observe minimum wage increases in some places and not others"

We can't control what we observe, but we can still try to use this variation to answer interesting causal questions!

Observational studies

Let's use the minimum wage increase in NJ as our running example.

1992, NJ increased minimum wage from \$4.25 to \$5.05 per hour

Causal question: Did this increase **reduce** or **increase** employment?

Several approaches:

Compare outcome in NJ to PA: cross-sectional comparison

Compare similar groups in NJ and PA: subclassification / statistical control

Compare same group before and after: panel comparison / before-after

Compare before-after in NJ to before-after in PA: difference-in-differences

Cross-sectional comparison

Define $\overline{Y}^{\text{NJ}}$ and $\overline{Y}^{\text{PA}}$ as average % full-time employees in fast-food restaurants.

$$\text{causal effect} = \overline{Y}^{\text{NJ}} \text{ (treatment)} - \overline{Y}^{\text{PA}} \text{ (control)}$$

Q: What are some issues?

Define $\overline{Y}^{\text{NJ}}$ and $\overline{Y}^{\text{PA}}$ as average % full-time employees in fast-food restaurants.

$$\text{causal effect} = \overline{Y}^{\text{NJ}}\ (\text{treatment}) - \overline{Y}^{\text{PA}}\ (\text{control})$$

Q: What are some issues?

Confounding! NJ and PA might differ in economies, politics, populations, etc.

Pick covariates X that might be confounding. Compute the causal effect only on units with **identical** values of X.

$$\text{causal effect} = \overline{Y}^{\text{NJ}}_{X=\text{Taco Bell}} - \overline{Y}^{\text{PA}}_{X=\text{Taco Bell}}$$

Matching: compare apples-to-apples!

Q: What other X's could be used?

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Q: What other X's could be used?

Geographic location, closeness to NJ-PA border, urban or rural area

What if we have data measured *over time*? Panel or longitudinal data.

$$Y^{\text{NJ}}_{i,\text{before}} \longrightarrow \textbf{min wage hike} \longrightarrow Y^{\text{NJ}}_{i,\text{after}}$$
$$Y^{\text{PA}}_{i,\text{before}} \longrightarrow \textbf{min wage hike} \longrightarrow Y^{\text{PA}}_{i,\text{after}}$$

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$$\text{causal effect} = \overline{Y}^{\text{NJ}}_{\text{after}} - \overline{Y}^{\text{NJ}}_{\text{before}}$$

Q: What are some issues?

Confounding again! Time is potentially an issue (remember ice cream).

Remove time-trend confounding by using panel data from both treated (NJ) and control (PA).

Four outcomes: $\overline{Y}^{\text{NJ}}_{\text{before}}, \overline{Y}^{\text{NJ}}_{\text{after}}, \overline{Y}^{\text{PA}}_{\text{before}}, \overline{Y}^{\text{PA}}_{\text{after}}$

$$\text{DiD} = (\overline{Y}^{\text{NJ}}_{\text{after}} - \overline{Y}^{\text{NJ}}_{\text{before}}) - (\overline{Y}^{\text{PA}}_{\text{after}} - \overline{Y}^{\text{PA}}_{\text{before}})$$

Why does this work?

answer 1: picture

answer 2: some math

Difference-in-differences (math explanation)

$4 \times N$ potential outcomes, 2 for each state: $\{Y_i^{\mathsf{NJ}}(0), Y_i^{\mathsf{NJ}}(1), Y_i^{\mathsf{PA}}(0), Y_i^{\mathsf{PA}}(1)\}$ for all i

$$
\begin{aligned}
\text{DiD} &= (\overline{Y}^{\mathsf{NJ}}_{\mathsf{after}} - \overline{Y}^{\mathsf{NJ}}_{\mathsf{before}}) - (\overline{Y}^{\mathsf{PA}}_{\mathsf{after}} - \overline{Y}^{\mathsf{PA}}_{\mathsf{before}}) \\
&= \Big(\mathbb{E}[Y_i^{\mathsf{NJ}}(1)|\mathsf{after}] - \mathbb{E}[Y_i^{\mathsf{NJ}}(0)|\mathsf{before}]\Big) \\
&\quad - \Big(\mathbb{E}[Y_i^{\mathsf{PA}}(0)|\mathsf{after}] - \mathbb{E}[Y_i^{\mathsf{PA}}(0)|\mathsf{before}]\Big) + 0 \\
&= \Big(\mathbb{E}[Y_i^{\mathsf{NJ}}(1)|\mathsf{after}] - \mathbb{E}[Y_i^{\mathsf{NJ}}(0)|\mathsf{after}]\Big) \\
&\quad + \Big(\mathbb{E}[Y_i^{\mathsf{NJ}}(0)|\mathsf{after}] - \mathbb{E}[Y_i^{\mathsf{NJ}}(0)|\mathsf{before}]\Big) \\
&\quad - \Big(\mathbb{E}[Y_i^{\mathsf{PA}}(0)|\mathsf{after}] - \mathbb{E}[Y_i^{\mathsf{PA}}(0)|\mathsf{before}]\Big)
\end{aligned}
$$

Difference-in-differences (math explanation)

$$\begin{aligned}\text{DiD} = &\Big(\mathbb{E}[Y_i^{\text{NJ}}(1)|\text{after}] - \mathbb{E}[Y_i^{\text{NJ}}(0)|\text{after}]\Big) \\ &+ \Big(\mathbb{E}[Y_i^{\text{NJ}}(0)|\text{after}] - \mathbb{E}[Y_i^{\text{NJ}}(0)|\text{before}]\Big) \\ &- \Big(\mathbb{E}[Y_i^{\text{PA}}(0)|\text{after}] - \mathbb{E}[Y_i^{\text{PA}}(0)|\text{before}]\Big)\end{aligned}$$

First part: $\mathbb{E}[Y_i^{\text{NJ}}(1)|\text{after}] - \mathbb{E}[Y_i^{\text{NJ}}(0)|\text{after}]$ = sample average treatment effect (of treated)

Second part: (slope of NJ line) − (slope of PA line)

Parallel trend assumption: second part $= 0$

The DiD estimate can be obtained from a simple **regression**. Suppose we have panel data with units i observed at times t:

$$Y_{it} = \alpha + \beta_1 \cdot \text{Treat}_i + \beta_2 \cdot \text{Post}_t + \delta \cdot (\text{Treat}_i \times \text{Post}_t) + \varepsilon_{it}$$

$\text{Treat}_i = 1$ if unit i is in the treatment group (e.g., NJ)

$\text{Post}_t = 1$ if time t is after the intervention

δ is the **DiD estimator** — this is the causal effect!